CARED-VR: Context-Aware Reinforcement and Enforcement for Drill Training in VR

Md Asif Bin Karim*
University of Arkansas at Little Rock

Karun Shrestha†
University of Arkansas at Little Rock

Atit Kharel‡
University of Arkansas at Little Rock

M. J. M. Sunny§
University of Arkansas at Little Rock

Aryabrata Basu¶
University of Arkansas at Little Rock

2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW) | 979-8-3195-0529-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/VRW70859.2026.00159

Abstract

Effective VR safety training needs context-aware regulation during task execution, not just immersive visuals or instructional feedback. We present CARED-VR, a context-aware VR training prototype that integrates a vision-language model (VLM) with haptic gloves to deliver graded reinforcement and enforceable safety constraints in a simulated power-drilling task. The VLM functions as a runtime supervisory layer that evaluates visually observable configurations, including drill-bit and material compatibility and ongoing tool use, against predefined safety conditions. Safe actions proceed uninterrupted, while unsafe configurations trigger a structured escalation of interventions: verbal guidance, simulated physical consequences, and haptic enforcement that physically prevents unsafe operation. We developed CARED-VR, a simulated workshop in which users perform drilling tasks on different materials within a virtual environment (VE) using the SenseGlove Nova 1 in conjunction with a Valve Index headset to enable immersive, haptically augmented interaction. The system illustrates how high-level contextual reasoning can be coupled with low-level embodied interaction to move beyond passive simulation toward execution-time safety control. In addition, CARED-VR serves as a research platform for examining user compliance, workload, and perceived realism.

Index Terms: VR training; Context-aware interaction; Vision-language model; Safety enforcement; Embodied interaction; Industrial tool simulation

1 Introduction

Virtual reality (VR) has become a practical medium for training workers to operate complex and potentially dangerous equipment, as it enables repeatable, high-fidelity scenarios without exposing trainees to real-world hazards [6, 16]. Industrial and construction domains increasingly rely on VR to rehearse tasks such as machine operation, working at height, and handling electrical systems, reporting improvements in safety awareness, engagement, and knowledge retention compared to traditional classroom or desktop-based instruction [5].

From an XRIOS perspective, however, the core challenge extends beyond realism. Effective training systems must provide scalable occupational support, including timely, context-sensitive feedback that prevents repeated unsafe actions while reinforcing correct technique and maintaining task flow. Many existing VR training systems fall short of this goal [13, 32]. They primarily emphasize visual and auditory fidelity, offering limited physical consequences for unsafe actions and little explicit guidance on fine-grained tool handling practices. Haptic interfaces offer a partial solution by reintroducing physical cues that are central to real-world equipment use, such as contact forces, tool resistance, and thermal feedback [30]. Prior work has shown that body-worn and hand-worn haptic devices can reinforce correct procedures by delivering immediate tactile responses to user errors and supporting muscle memory formation in safety-critical tasks. Devices such as the SenseGlove Nova combine force feedback, vibrotactile actuation, and precise finger tracking, and have been successfully deployed in domains ranging from industrial tool training to surgical simulation, where users must manipulate virtual instruments with realistic resistance and contact sensations [2]. These advances move VR training closer to the embodied demands of real work [26]. However, haptics alone does not address the need for context-aware supervision, specifically, deciding when feedback should escalate from guidance to consequence.



Figure 1: CARED-VR simulated workshop environment for VR-based power-tool safety training and intervention. The environment allows users to perform drilling tasks on different virtual materials using immersive, haptically augmented interaction.

In parallel, large language models (LLMs) have emerged as flexible, domain-adaptable agents capable of interpreting contextual information and providing natural-language feedback about safety-related behavior [24, 34]. Recent research has explored LLMs as "virtual safety officers" that answer domain-specific questions or personalize construction safety training based on user actions and scene context [4, 15, 24]. While promising, most LLM-based safety systems operate at the level of static question–answer interaction or coarse scenario events. They are rarely coupled to continuous physical interaction or to mechanisms that can enforce safety constraints in runtime. This limitation is critical, as many industrial accidents arise not from a single catastrophic error, but from small, repeated deviations in setup and technique.

This paper introduces a VLM-driven haptic training system for safe handheld power-drill operation in VR that closes the loop between context-aware safety supervision and low-level physical interaction. The system is implemented in a Unity-based virtual workshop (cf. Figure 1) featuring a SenseGlove Nova–controlled drill,

*e-mail: mkarim1@ualr.edu
†e-mail: kshrestha@ualr.edu
‡e-mail: akharel@ualr.edu
§e-mail: msunny@ualr.edu
¶e-mail: abasu@ualr.edu

979-8-3195-0529-3/26/$31.00 ©2026 IEEE
DOI 10.1109/VRW70859.2026.00159

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

interchangeable bits, physics-based material carving, and a simplified thermal model. A vision-language model (VLM) periodically analyzes rendered first-person views together with task context to assess drill bit–material compatibility and ongoing tool use.
Rather than treating the AI as a conversational assistant, the model functions as a decision layer that governs intervention escalation. Correct actions proceed without interruption. When unsafe configurations or behaviors are detected, the system escalates feedback in a structured manner: verbal reinforcement, simulated physical consequences such as visual overheating and tool stall, and finally, enforceable haptic constraints that physically prevent continued operation. Concretely, the drill bit visibly heats, smoke effects intensify, the motor stalls, and the SenseGlove enforces an index-finger lock that blocks further trigger pulls until the user selects an appropriate bit or allows the tool to cool.

Scope of safety supervision. In this paper, "unsafe" refers specifically to configuration-level misuse that can be observed in the virtual scene and operational state: primarily drill-bit/material incompatibility and continued operation under an incompatible configuration. The current prototype does not model broader real-world drilling hazards *(e.g., PPE compliance, stance/posture, kickback)*, and we therefore focus on demonstrating an execution-time supervision and enforcement loop within a tightly defined task scope.

Novelty and scope. The novelty of CARED-VR is not the use of a haptic glove or a vision-language model in isolation, but a design pattern for VR occupational training in which (i) safety is defined as a bounded, auditable set of unsafe configurations, (ii) a runtime supervisory layer assesses these configurations during execution using perceptual evidence plus structured tool state, and (iii) enforcement is gated by conservative consistency checks and staged escalation, with a reversible haptic lock as a last resort. This shifts VR safety training from post-hoc instruction to execution-time regulation of unsafe behavior.

The contributions of this work are threefold:

- A context-aware safety intervention workflow that integrates VLM-based scene assessment with continuous, embodied interaction for a VR task.
- How haptic gloves can function not only as a realism and feedback channel, but also as enforcement mechanism that physically blocks impermissible actions (trigger pulls) under unsafe configuration while keeping recovery user-driven.
- A prototype research platform for initial technical validation on how graded intervention strategies, ranging from warnings to physical lockouts, affect compliance, learning, workload, and perceived realism in haptics-rich VR safety training.

2 Related Work

2.1 VR Training for Safety-Critical Tasks

Virtual reality (VR) has been widely adopted for training in safety-critical domains due to its ability to provide immersive, repeatable, and risk-free environments for practicing hazardous procedures. Prior work has demonstrated the effectiveness of VR training for industrial operations, construction safety, and equipment handling, reporting improvements in engagement, situational awareness, and procedural understanding compared to traditional instructional methods [7, 13, 25].

Several systems have explored VR-based simulations for power-tool operation, machining, and assembly tasks, often emphasizing visual realism and task flow [3, 22]. However, many of these approaches rely on scripted logic or static rule sets, offering limited adaptability to nuanced or repeated unsafe behavior during continuous interaction.

Recent reviews of VR safety training highlight that many systems lack consistent evaluation protocols and that long term retention and theory grounded design are often missing, motivating prototype papers that report auditable scope and measurable loop behavior [18]. Complementary evidence from immersive procedural training reviews emphasizes wide variability in interventions and outcomes, supporting the need to report concrete implementation details and engineering level performance characteristics alongside user facing measures [9].

2.2 Context-Aware Systems in VR and XR

Context awareness has long been recognized as a key requirement for adaptive interactive systems, including VR and broader extended reality (XR) applications. Established taxonomies define context in terms of human state, environmental conditions, and system status, enabling applications to adapt behavior and content based on situational constraints [1, 17]. In immersive systems, context awareness has been used to adapt instructional content, visualization placement, and task progression [21].

Recent work has explored context-aware authoring workflows that ground generated content in spatial and temporal information. CARING-AI exemplifies this direction by enabling authors to create context-aware augmented reality instructions using generative AI, capturing human, environmental, and system context during authoring to situate avatar-based demonstrations [20]. These systems primarily focus on contextualizing instructional content prior to execution.

2.3 Vision–Language Models for Safety and Training in VR and XR

Large language models (LLMs) and vision–language models (VLMs) have recently been introduced into VR and XR systems as flexible reasoning components capable of interpreting task context and generating adaptive feedback [23, 33]. Prior work has explored these models as conversational safety advisors or adaptive instructors that provide explanations and recommendations based on scene descriptions or symbolic task representations [12].

However, most AI-enhanced safety systems operate at the dialog or narrative level and remain loosely coupled to continuous physical interaction. AI-driven feedback is typically advisory, with limited authority to influence or constrain ongoing user actions. Recent surveys have identified the lack of grounding between high-level AI reasoning and low-level embodied interaction as a key challenge for deploying generative models in immersive systems [8, 12]. In contrast to prior VR and XR systems that use AI primarily for instruction or explanation, our work embeds a VLM directly into the interaction loop as a supervisory decision layer.

2.4 Haptic Gloves for Multimodal Feedback and Enforcement in VR

Haptic interfaces, including force-feedback gloves, are commonly used in VR to enhance realism and support skill acquisition by providing tactile and kinesthetic cues during object manipulation [11, 14]. In training contexts, haptic feedback has been shown to improve fine motor control and reinforce correct procedures [28].

Commercial devices such as the SenseGlove Nova enable per-finger force feedback and vibrotactile actuation, making them suitable for industrial training and tool-based interaction [19]. Nevertheless, in most existing VR systems, haptics function as feedback mechanisms rather than enforcement tools; unsafe actions may be signaled but are rarely physically prevented. Our work extends prior VR haptics research by treating force feedback not only as a feedback channel, but as a mechanism for enforcing safety constraints during ongoing interaction.

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

2.5 From Context-Aware Instruction to Runtime Enforcement

Taken together, prior work in VR training, context-aware XR systems, generative AI, and haptics highlights a gap between adaptive instruction and active behavior regulation [12–14]. Context-aware systems such as CARING-AI demonstrate how generative models can ground instructional content in human and environmental context, but primarily operate at authoring time [20]. Conversely, haptic VR training systems provide embodied feedback but lack high-level contextual reasoning. This work bridges these directions by combining runtime context evaluation with haptic enforcement, positioning AI-driven supervision as a mechanism for regulating behavior during task execution rather than guiding interpretation alone.

3 CARED-VR System

3.1 Hardware and Tracking Setup

The system runs in Unity 2022.3 using the Universal Render Pipeline (URP), targeting a PC VR configuration with an OpenVR/OpenXR-compatible head-mounted display and a pair of SenseGlove Nova haptic gloves. The Nova gloves provide wireless force feedback through per-finger locking mechanisms and vibrotactile actuators on the thumb, index finger, and palm, enabling both gross resistance *(e.g., trigger pull)* and fine-grained vibration cues during drilling [29].

The virtual hands are implemented as layered SenseGlove's Hand prefabs, with separate sublayers for visual skinned meshes, physics colliders, grab detection, gesture recognition, and haptic feedback. Each hand maintains a distinction between a "real" tracked pose *(from the glove and HMD)* and a "virtual" interaction pose that can be slightly offset or constrained to ensure robust object manipulation while still feeling physically plausible. This architecture allows the system to impose haptic constraints–such as a force-locked index finger–without visually breaking hand–object alignment, which is critical for preserving presence in VR training tasks.

The drill trigger is actuated from the SenseGlove index-finger flexion value ($\in [0, 1]$), mapped to a continuous trigger value and discretized into a "locked" state using a fixed threshold (*set at flexion value 0.5*). During *enforcement* (Sec. 3.5), the system commands the Nova's index-finger locking channel to apply a sustained resistive force (*approximately 20N per finger*) that prevents further flexion beyond the engagement threshold; the lock is released immediately once the unsafe condition is cleared [19]. This implementation treats enforcement as an input constraint *(blocking trigger engagement)* rather than a change in hand pose tracking, preserving visual hand–tool alignment.

3.2 Tracking and Alignment

We align the SenseGlove hand pose to the VR hand model using the vendor-provided calibration and a fixed transform between the glove coordinate frame and the virtual hand rig. To sanity-check alignment for drilling, we perform a repeatability check in which users place the drill tip on marked targets in the workspace; we log tip-to-target offset and verify that offsets remain within an acceptable tolerance for consistent contact and deformation behavior.

3.3 Virtual Workshop and Power-Drill Model

The training scenario is situated in a fully virtual workshop that represents a compact industrial drilling environment. The scene includes three representative workpieces, wood, concrete, and metal, which serve as material categories for the drilling task. These materials were selected to reflect common industrial conditions while allowing clear visual and functional differentiation during interaction (cf. Figure 3). The spatial layout is designed to support natural reach, grasp, and alignment, providing sufficient visual context for tool handling and material recognition.



(a) (b) (c)

Figure 2: Drill bit types used in the VR training task: (a) brad point bit for wood, (b) masonry bit for concrete, and (c) high-speed steel (HSS) bit for metal.

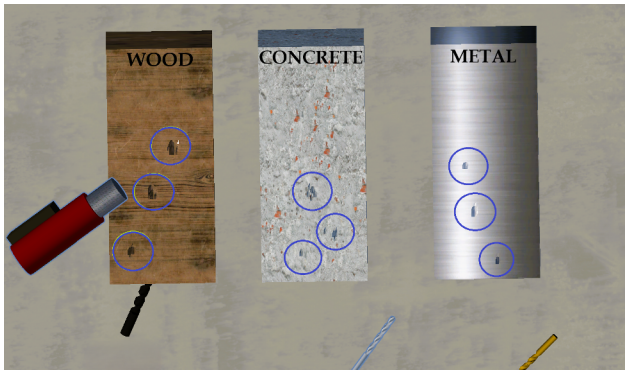


Figure 3: Virtual workshop scene illustrating three material slabs, wood, concrete, and metal, used as surface categories in the drilling task and the drilled mark in the surface using compatible bit.

The primary interactive object is a handheld power drill operated using the SenseGlove Nova haptic gloves. The drill model incorporates a rigged trigger, rotating chuck, physics colliders, and synchronized visual, auditory, and haptic feedback. Index-finger flexion controls trigger engagement, while force and vibrotactile feedback convey operational resistance, contact, and tool state during drilling.

To emphasize correct tool selection as a safety-critical factor, the system models three material-appropriate drill bit types commonly used in industrial practice: brad point bits for wood, masonry bits for concrete, and high-speed steel (HSS) twist bits for metal drilling [31]. Each drill bit is implemented as a distinct, grabbable object that can be physically mounted into the drill chuck using direct hand interaction.

Whenever the user grasps a drill bit, an informational overlay is displayed near the hand, presenting key properties of the selected bit, including diameter, length, pitch, and material compatibility (cf. Figure 2). This contextual information is shown at the moment of interaction, allowing users to verify bit suitability before mounting it into the drill. The overlay is non-intrusive and disappears once the bit is released or attached, ensuring that guidance is available without interrupting task flow.

Once attached, the active drill bit configures interaction parameters such as resistance, vibration intensity, thermal response, and visual effects. Appropriate bit–material pairings result in stable drilling behavior and consistent haptic feedback, while incompatible combinations lead to degraded performance, accelerated visual heating, increased vibration irregularity, and eventual tool stall. These responses establish a direct and observable relationship between tool selection, material properties, and drilling outcomes, reinforcing correct configuration through embodied interaction rather than explicit

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

instruction.

Hole Formation To preserve a clear visual consequence of drilling without heavy computation, we approximate hole formation by locally displacing mesh vertices around the drill tip when contact is detected. The deformation depth is proportional to penetration and advances along the drill axis, producing a stable, real time cavity that is sufficient for guiding tool use and supporting the safety interventions described later.

3.4 Vision–Language–Based Task Assessment

User drilling behavior is analyzed within a Unity-based VR training environment through periodic, task-level assessment using a vision–language model (VLM). Rather than operating on continuous video streaming, the system performs repeated frame-based inference during task execution. This design reflects the observation that safety-critical drilling errors, such as incorrect bit selection or sustained misuse, typically persist over multiple seconds rather than occurring instantaneously.

The virtual scene is rendered at a resolution of 854×480 pixels, which preserves sufficient semantic detail for material and tool-state interpretation while supporting efficient transmission and low-latency inference. Every three seconds, Unity captures a first-person frame and packages it with auxiliary task metadata describing the current drilling configuration. These metadata include the currently attached drill bit type, drill's rotational speed (RPM), and a binary interaction state indicating whether the drill bit is in contact with a workpiece or not. Providing these signals alongside visual input reduces ambiguity that may arise from appearance alone, such as distinguishing visually similar drill bits or inferring whether drilling is actively taking place.

Each captured frame is transmitted to a self-hosted VLM endpoint together with a structured prompt encoding task rules and material–bit compatibility constraints (e.g., HSS bit for metal, brad-point bit for wood, masonry bit for concrete). The VLM reasons jointly over the visual scene and the accompanying metadata to infer the material being drilled and to assess whether the observed action aligns with expected drilling procedure. The model produces a compact safety assessment rather than free-form instruction, indicating whether the current configuration is permissible or unsafe. Typical inference latency ranges from 1.5 to 2.0 seconds.

To improve reliability and reduce the risk of hallucinated or internally inconsistent judgments, the VLM output is evaluated by a secondary verification stage implemented using a language-only model. This verifier examines the alignment between the VLM's inferred material, its stated reasoning, and the provided task metadata, identifying contradictions or unsupported inferences through few-shot prompting. Reinforcement is applied only when both stages independently conclude that the configuration violates task constraints. Outputs that fail this consistency check are treated as a no-decision outcome and ignored, ensuring that the system does not issue ambiguous or potentially misleading guidance to the user.

In our implementation, we employ the Qwen2.5-VL-7B vision–language model as the perceptual backbone for task assessment and the Qwen2.5-7B large-language model as a secondary verifier for consistency [10]. This model was selected due to its strong visual grounding capabilities, moderate parameter scale, and compatibility with local deployment. At approximately 7B parameters, Qwen2.5-VL-7B provides sufficient semantic reasoning to distinguish tool geometry and material appearance while maintaining inference latency suitable for interactive XR training scenarios. Importantly, its instruction-following behavior enables the generation of structured, machine-readable outputs, which integrate directly with the downstream reinforcement and enforcement logic. While the present work focuses on system integration rather than model comparison, the CARED-VR pipeline is model-agnostic and can accommodate alternative vision-language models with similar multimodal reasoning capabilities.

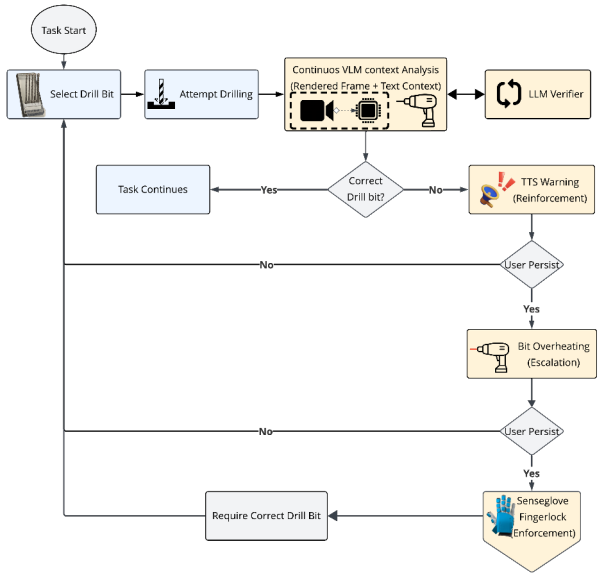


Figure 4: CARED-VR safety supervision workflow. A vision–language model analyzes rendered first-person views to assess drill bit–material compatibility. Correct actions proceed without interruption. Incorrect actions trigger graduated interventions: verbal reinforcement, simulated physical consequences, and haptic enforcement that physically prevents continued unsafe operation.

The three-second sampling interval was selected empirically as a balance between responsiveness and computational efficiency. This interval provides timely supervision for sustained unsafe behavior while avoiding the noise sensitivity and overhead associated with frame-by-frame analysis.

3.5 User Interaction and Safety Supervision Logic

User interaction is organized as a continuous control loop governed by task state, visual context, and enforcement state. After calibration, the user enters the virtual workshop, grasps the drill, selects a drill bit using hand in virtual environment, and mounts it into the chuck through direct hand interaction. The drill is then positioned against a target material, and index-finger flexion is mapped to trigger engagement, activating drilling with synchronized visual, auditory, and haptic feedback.

The overall safety supervision workflow is illustrated in Figure 4. During operation, the system periodically monitors user actions and tool state using a vision–language model that analyzes rendered first-person views together with task context, including drill bit type, target material, and tool activity. This evaluation runs implicitly at each interaction step and does not interrupt normal task execution when conditions are safe.

When the evaluated configuration satisfies compatibility and safety conditions, drilling proceeds without interruption, and the task state advances normally. In this case, supervision remains transparent to the user, preserving natural task flow and immersion. If an incompatible drill bit–material pairing or unsafe operating behavior is detected, the system transitions to a reinforcement state, as formalized in Algorithm 1. In this state, a brief verbal warning is issued to signal the detected condition while preserving uninterrupted physical interaction.

If unsafe behavior persists after reinforcement, the system esca-

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 CARED-VR : Context-Aware Safety Supervision

INPUT: F_t ▷ Rendered first-person frame at time t
C_t ▷ Task context (bit type, material, tool state)
OUTPUT: S_t ▷ Supervision state

Initialize supervision state $S_t \leftarrow$ NORMAL
while task active **do**
  Capture rendered frame F_t
  Retrieve task context C_t
  Evaluate bit–material compatibility using VLM
  if configuration is safe **then**
    $S_t \leftarrow$ NORMAL ▷ Task continues uninterrupted
  else
    if $S_t =$ NORMAL **then**
      Issue verbal warning
      $S_t \leftarrow$ REINFORCEMENT
    else if $S_t =$ REINFORCEMENT **then**
      Apply simulated consequences
        (visual heating, smoke, drill stall)
      $S_t \leftarrow$ ESCALATION
    else if $S_t =$ ESCALATION **then**
      Activate SenseGlove Index finger lock
      Block trigger input
      $S_t \leftarrow$ ENFORCEMENT
    end if
  end if
  if $S_t =$ ENFORCEMENT **and** correct bit selected **then**
    Release Index finger lock
    $S_t \leftarrow$ NORMAL
  end if
end while

lates to a consequence state. In this state, continued drilling with an incompatible bit–material pairing *(e.g., attempting to drill metal with a brad-point bit, or wood with a masonry bit)* produces visible and progressive simulated misuse effects. Most prominently, the active drill bit begins to glow red, indicating localized overheating due to improper configuration. This visual heating is accompanied by smoke emission and increasing rotational instability, culminating in partial drill stall as illustrated in Figure 5. These consequences are designed to mirror failure signatures of improper drilling while remaining fully reversible, enabling recovery through corrective action such as selecting a material-appropriate bit or disengaging the tool.

Continued operation under unsafe conditions triggers the enforcement state. As shown in Figure 4 and specified in Algorithm 1, the SenseGlove Nova applies force feedback to physically constrain the user's index finger (Sec. 3.1), preventing further trigger pulls and immediately halting drilling. Enforcement remains active until the unsafe condition is resolved, either by selecting a material-appropriate drill bit or by allowing the simulated tool state to recover.

Once the unsafe condition is resolved, enforcement is released and the system returns to the normal operation state, allowing the task to continue. Across all states, safety supervision operates periodically and implicitly, without requiring explicit confirmations or menu-based input. Corrective actions are performed through embodied manipulation rather than symbolic interaction, embedding safety constraints directly into the mechanics of tool use and reinforcing the causal relationship between configuration, action, and consequence.

3.6 Design Rationale and Scope of Haptic Enforcement

A potential concern with the proposed system is the use of haptic enforcement to physically constrain user actions, which may be perceived as overly restrictive or disruptive to user agency. We emphasize that enforcement in CARED-VR is intentionally designed as a *last-resort intervention* within a graduated supervision framework rather than a default response. As shown in Figure 4 and formalized in Algorithm 1, enforcement is only activated after repeated unsafe behavior persists despite prior reinforcement and simulated physical consequences.

This design reflects the realities of safety-critical tool operation, where certain actions are not merely inadvisable but impermissible. In real-world drilling tasks, continued misuse of a tool under unsafe configurations can lead to irreversible damage or injury, at which point corrective intervention is no longer optional. By physically constraining trigger engagement through force feedback, the system enforces a clear boundary between permissible and impermissible actions, mirroring real-world safety constraints rather than replacing them with abstract warnings.

Importantly, enforcement does not remove user agency but instead redirects it. Users retain control over task recovery by selecting an appropriate drill bit or allowing the simulated tool state to recover. Enforcement is therefore corrective rather than punitive, guiding users back to a safe operational state through embodied action rather than symbolic confirmation or system-driven reset. This approach preserves immersion and task continuity while ensuring that repeated unsafe behavior cannot continue unchecked.

Finally, we note that the proposed system is intended as a research platform rather than a prescriptive training solution. The enforcement logic is deliberately explicit and transparent, enabling future work to systematically study how different intervention thresholds, enforcement durations, and recovery conditions affect compliance, learning, workload, and perceived realism. By making enforcement an explicit design choice rather than an implicit side effect, CARED-VR provides a controlled foundation for investigating the role of embodied constraint in context-aware VR safety training.

Req.	Requirement	Implementation in CARED-VR
R1	Context awareness during execution (bit, material, tool state)	Runtime supervisory assessment over periodic frames plus structured tool-state metadata.
R2	Auditable safety scope (bounded unsafe set)	Predefined unsafe conditions (bit-material mismatch, overheating, overload) and staged escalation logic.
R3	Graded interventions before enforcement	Guidance and warnings precede simulated consequences; enforcement is last resort and reversible.
R4	Enforceable stoppage that is reversible	SenseGlove index-finger force lock released on corrective action or hazard resolution.
R5	Low overhead instrumentation and monitoring	Lightweight event logging and dashboards for inspection and replay of unsafe episodes.

Table 1: Design requirements and traceability to implementation.

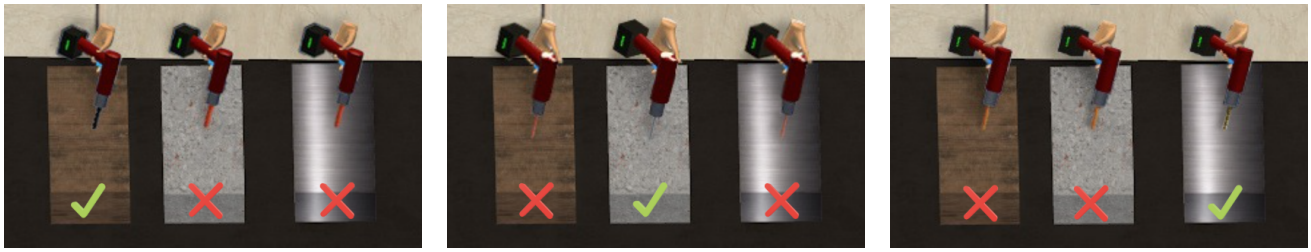
3.7 Design Goals and Requirements

We derived the following requirements from the target occupational training setting (Table 1) of handheld power drilling, where unsafe outcomes often arise from small deviations in setup and technique that repeat during execution. Our goal is not to certify real world competence, but to provide scalable occupational support during VR practice by detecting bounded safety relevant misconfigurations and applying graded interventions that are interpretable and reversible.

3.8 Development Methodology

We followed an iterative prototype driven process. We first implemented the embodied drilling interaction and hazard state models, then introduced instrumentation to observe failures during repeated task execution, and finally integrated the VLM supervisory layer

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Figure 5: Bit selection outcomes across materials. (a) Brad-point bit for wood. (b) Masonry bit for concrete. (c) HSS bit for metal. In incompatible configurations, the system flags risk by turning the bit emissive red and initiating the staged intervention sequence (warning, stall, and if persistent, haptic lock).

with conservative gating and escalation rules. Each iteration focused on reducing ambiguity in what the system detects, limiting enforcement to bounded unsafe cases, and ensuring that interventions remain reversible and do not introduce new discomfort risks. Our iterative prototype development focused on satisfying R1–R5 (Table 1) while keeping enforcement reversible.

4 Engineering Evaluation

We report an engineering evaluation to characterize whether the supervisory loop is usable for execution time safety regulation.

4.1 Latency and Cadence

The system samples one frame every 3 seconds at 854×480 resolution and performs VLM inference with a typical end to end latency of 1.5–2.0 seconds from frame capture to intervention decision. We log timestamps for capture, inference start, inference completion, and intervention actuation to enable replication and future optimization.

4.2 Scenario Based Reliability Check

We constructed a small scripted set of safety critical scenarios that exercise the bounded unsafe conditions (bit material mismatch, overheating, overload). For each scenario we record whether the system (i) detects the unsafe state, (ii) escalates in the intended order, and (iii) releases enforcement after correction or hazard resolution. We treat VLM verifier disagreement as "no decision" and require persistence across samples before haptic enforcement.

4.3 Engineering Metrics

We instrumented the supervisory loop to log frame capture time, inference start and end, and intervention actuation. In our local deployment, the loop samples one frame every 3 seconds and produces a decision with typical end-to-end latency of 1.5–2.0 seconds. Across sampled trials, we report median and p95 latency and the fraction of samples discarded as *no decision* when the verifier disagrees, since enforcement is only permitted after consistent detections.

5 Discussion and Limitations

This work examines how vision–language models can be integrated into haptics-rich VR training as supervisory safety agents that regulate unsafe behavior during task execution rather than providing conversational instruction. By embedding a VLM directly into the interaction loop of a power-drilling task, CARED-VR demonstrates how visual context evaluation can drive graduated safety intervention, aligning with the need to regulate unsafe actions as they occur rather than addressing them post hoc.

Rather than positioning the AI component as a conversational assistant, CARED-VR frames the vision–language model as a runtime decision layer that evaluates visually observable task configurations, including drill bit–material compatibility and ongoing tool use, against predefined safety conditions. When configurations are safe, interaction proceeds uninterrupted. When unsafe conditions are detected, the system escalates intervention in a structured manner, progressing from verbal reinforcement to simulated physical consequences and, when necessary, haptic enforcement that prevents continued unsafe operation.

A key design contribution of this work is the explicit use of haptic feedback as an enforcement mechanism rather than solely as a feedback channel. By embedding safety constraints into the mechanics of tool use, the system moves beyond passive simulation toward active regulation of behavior while preserving user agency through reversible, execution-time intervention. Enforcement is applied selectively and proportionally, ensuring that users remain responsible for resolving unsafe configurations through embodied action rather than system-driven correction.

Several limitations define the scope of the present work. First, CARED-VR is implemented as a conceptual prototype with a limited set of materials, drill bits, and visually observable safety cues. The supervision logic operates on rendered first-person views and predefined task context, and does not claim generalization beyond the modeled drilling scenarios. Extending this approach to more complex tools, materials, or failure modes would require additional domain modeling and validation.

Second, this work does not evaluate training effectiveness, skill transfer [27], or long-term learning outcomes. While the system is designed to regulate unsafe behavior through embodied enforcement, no claims are made regarding improvements in real-world performance. Controlled user studies are required to examine how intervention thresholds, escalation timing, and enforcement duration influence compliance, workload, learning, and perceived realism.

Third, although haptic enforcement operates entirely within the virtual environment and is designed to be perceptible yet non-painful, individual differences in sensitivity and tolerance may affect user experience. Future work should investigate adaptive enforcement strategies that tailor force magnitude or modality to individual users while maintaining consistent safety guarantees.

The current implementation also assumes a single-user training scenario with a centralized supervisory agent. Collaborative training settings, instructor override mechanisms, and adaptive supervision policies remain open research directions and will be important for extending context-aware enforcement to more realistic instructional environments.

Failure Modes and Mitigations. Enforcement-based supervision introduces the risk of false positives that could prematurely restrict user action. CARED-VR mitigates this risk through staged intervention and temporal persistence: haptic enforcement is only activated when unsafe conditions are detected consistently across multiple evaluation steps and persist despite prior verbal reinforcement and simulated physical consequences. This design reduces sensitivity to transient perception errors or brief exploratory actions. Enforcement remains reversible and bounded; users can immediately

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

recover by correcting the configuration or allowing the simulated tool state to stabilize, and no irreversible penalties are applied. To address discomfort risk, the finger lock is implemented using limited braking forces within the operating range of the haptic glove and is released as soon as safe conditions are restored. By escalating intervention gradually rather than abruptly, the system also aims to manage workload and minimize startle effects, ensuring that supervision supports safe task execution without introducing unnecessary cognitive or physical burden.

Taken together, these findings position CARED-VR as a prototype research platform for investigating how context-aware AI supervision, physical simulation, and haptic constraint can operate as a unified system. By treating enforcement as an explicit and proportional design choice rather than an implicit side effect, this work provides a controlled foundation for studying how embodied constraint shapes compliance, workload, and perceived realism in safety-critical VR training tasks.

5.1 Limitations and Next Validation Steps

This workshop paper reports a working prototype and does not claim training transfer or injury reduction. Next, we will conduct a formative study to quantify safety compliance (unsafe episode rate and duration), workload, and perceived intrusiveness under different intervention schedules, followed by a controlled comparison of guidance-only versus enforcement-enabled conditions.

6 Conclusion

This paper presented CARED-VR, a context-aware VR training prototype designed to address a central limitation of current immersive safety training systems: the lack of execution-time supervision that can regulate unsafe behavior as it occurs. Motivated by the need to move beyond realism and instructional feedback alone, CARED-VR integrates vision–language–based scene assessment with haptics-rich interaction to support timely, context-sensitive safety intervention during tool use.

By embedding a vision–language model directly into the interaction loop, the system evaluates visually observable task configurations, specifically drill bit–material compatibility and ongoing tool operation, and governs when intervention is required. Rather than functioning as a conversational or advisory agent, the model serves as a supervisory decision layer that enables graduated intervention. Safe actions proceed uninterrupted, while persistent unsafe configurations trigger a structured escalation from verbal reinforcement to simulated physical consequences and, when necessary, enforceable haptic constraint. In this way, safety boundaries are embedded into the mechanics of interaction, aligning feedback, consequence, and recovery with embodied user action.

Within its defined scope, CARED-VR demonstrates how haptic interfaces can function not only as channels for realism and feedback, but also as mechanisms for enforcing impermissible actions while preserving user agency through reversible, user-driven recovery. Although the current prototype focuses on a narrowly defined drilling task and configuration-level misuse, it establishes a concrete foundation for studying how context-aware supervision and embodied enforcement interact to shape compliance, workload, and perceived realism.

More broadly, this work outlines an execution-time safety paradigm for VR training in which contextual AI reasoning and physical interaction are tightly coupled to regulate behavior, rather than merely inform it. As a research platform, CARED-VR enables systematic exploration of graded intervention strategies in safety-critical immersive environments, contributing to the development of scalable, context-aware occupational support systems that prioritize safe task execution alongside realism and engagement.

References

[1] G. D. Abowd, A. K. Dey, P. J. Brown, N. Davies, M. Smith, and P. Steggles. Towards a better understanding of context and context-awareness. *Handheld and Ubiquitous Computing*, pp. 304–307, 1999. doi: 10.1007/3-540-48157-5_29

[2] J. Boutin, S. Kamoonpuri, et al. Smart haptic gloves for virtual reality surgery simulation: A pilot study on external ventricular drain training. *Frontiers in Robotics and AI*, 10:1273631, 2024. doi: 10.3389/frobt.2023.1273631

[3] Y. Chen, Y. Li, and X. Wang. Virtual reality simulation for drilling training in manufacturing. *Procedia Manufacturing*, 5:247–258, 2016. doi: 10.1016/j.promfg.2016.08.022

[4] V. Dudhee, P. D. Bandara, C. Psarros, and V. Vukovic. Artificial intelligence-enhanced virtual reality for health and safety training in construction. In *Proceedings of the 15th International Conference on eLearning (eLearning 2024)*, vol. 3938 of *CEUR Workshop Proceedings*, pp. 74–82. CEUR-WS, 2024.

[5] R. Eiris, M. Gheisari, and B. Esmaeili. Desktop-based safety training using 360-degree panorama and static virtual reality techniques: A comparative experimental study. *Automation in construction*, 109:102969, 2020.

[6] Z. Feng, R. Lovreglio, T. W. Yiu, D. M. Acosta, B. Sun, and N. Li. Immersive virtual reality training for excavation safety and hazard identification. *Smart and sustainable built environment*, 13(4):883–907, 2024.

[7] L. Freina and M. Ott. A literature review on immersive virtual reality in education. *eLearning & Software for Education*, 1:133–141, 2015.

[8] W. Huang et al. Grounding large language models with contextual memory for embodied agents. *arXiv preprint arXiv:2401.01234*, 2024.

[9] J. Jongbloed, R. Chaker, and E. Lavoué. Immersive procedural training in virtual reality: A systematic literature review. *Computers & Education*, 221:105124, 2024. doi: 10.1016/j.compedu.2024.105124

[10] W. Li and F. Zhang. Real-time vision–language analysis for autonomous underwater drones: A cloud–edge framework using qwen2.5-vl. *Drones*, 9(9):605, 2025.

[11] D. Liu, X. Tong, and Y. Zhang. A survey of haptic glove technologies. *IEEE Access*, 8:16941–16955, 2020. doi: 10.1109/ACCESS.2020.2968014

[12] U. Mittal, S. Sai, V. Chamola, and D. Sangwan. A comprehensive review on generative ai for education. *Ieee Access*, 12:142733–142759, 2024.

[13] A. Moreno, D. Röltgen, and C. M. Schlick. Virtual reality for training in the manufacturing industry: A systematic review. *Applied Sciences*, 9(22):4917, 2019. doi: 10.3390/app9224917

[14] C. Pacchierotti, S. Sinclair, M. Solazzi, and A. Frisoli. Wearable haptic systems for the fingertip and the hand: Taxonomy, review, and perspectives. *IEEE Transactions on Haptics*, 10(4):580–600, 2017. doi: 10.1109/TOH.2017.2689006

[15] C. Park et al. Personalized construction safety training system using conversational ai-based virtual reality. *Automation in Construction*, 2025. Forthcoming; see preprint SSRN 5043158 and ScienceDirect record S092658052500247X.

[16] S. Pastel, K. Petri, C.-H. Chen, A. M. Wiegand Cáceres, M. Stirnatis, C. Nübel, L. Schlotter, and K. Witte. Training in virtual reality enables learning of a complex sports movement. *Virtual Reality*, 27(2):523–540, 2023.

[17] B. Schilit, N. Adams, and R. Want. Context-aware computing applications. *Proceedings of the Workshop on Mobile Computing Systems and Applications*, pp. 85–90, 1994. doi: 10.1109/WMCSA.1994.16

[18] D. Scorgie, Z. Feng, D. Paes, F. Parisi, T. Yiu, and R. Lovreglio. Virtual reality for safety training: A systematic literature review and meta-analysis. *Safety Science*, 171:106372, 2024. doi: 10.1016/j.ssci.2023.106372

[19] SenseGlove. Senseglove nova: Force-feedback haptic glove. `https://senseglove.com`, 2021. Accessed 2025.

[20] J. Shi, R. Jain, S. Chi, H. Doh, H.-g. Chi, A. J. Quinn, and K. Ramani. Caring-ai: Towards authoring context-aware augmented reality instruction through generative artificial intelligence. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2025. doi: 10.

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.

1145/3706598.3713348
[21] B. J. Souza, A. L. Szejka, and R. Z. Freire. Towards an integration of augmented reality in industrial assembly processes: Issues and perspectives. *Procedia Computer Science*, 253:1063–1072, 2025.
[22] R. J. Stone. Virtual environments for machining training. *Virtual Reality*, 5(3):145–154, 2001. doi: 10.1007/s100550100014
[23] K. Subasinghe et al. Large language models as virtual safety officers in immersive environments. *IEEE Computer Graphics and Applications*, 2024.
[24] S. M. S. Subasinghe, S. G. Gersib, T. M. Frueh, and N. P. Mankad. Can large language models (llms) act as virtual safety officers? *ACS Chemical Health & Safety*, 2024. doi: 10.1021/acs.chas.4c00097
[25] M. J. M. Sunny and A. Basu. Non-linear parameterization of spatial decision making in immersive virtual environment. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 389–395, 2024. doi: 10.1109/VRW62533.2024.00076
[26] M. J. M. Sunny, J. P. Springer, and A. Basu. Combined effects of vr and gaming expertise on precision performance. *Human Factors in Design, Engineering, and Computing. AHFE (2025) International Conference. AHFE Open Access, vol 199. AHFE International, USA.*, 199, 2025. doi: 10.54941/ahfe1006907
[27] M. J. M. Sunny, J. P. Springer, and A. Basu. Evaluating adjustment and proficiency disparities in virtual reality. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1282–1283. IEEE, 2025.
[28] A. Talvas, M. Marchal, and A. Lecuyer. Haptic training for industrial assembly tasks. *IEEE Transactions on Haptics*, 7(3):333–346, 2014. doi: 10.1109/TOH.2014.2315797
[29] Unbound XR. Senseglove nova — haptic virtual reality gloves, 2023. Accessed: 2025-12-09.
[30] S. Villenave, P. Raimbaud, and G. Lavoué. Dynamic and modular thermal feedback for interactive 6dof vr: Implementation and evaluation. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1204–1205. IEEE, 2025.
[31] J. R. Walker. *Machining Fundamentals*. Goodheart-Willcox, 2nd ed., 2014.
[32] Y. Wang and L. Pan. Immersive virtual reality in education: impact on the emotional intelligence of university students. *Education and Information Technologies*, 30(4):5283–5299, 2025.
[33] R. Zhang et al. isafetrainer: Ai-driven adaptive safety training in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2025.
[34] Y. Zhang et al. A comprehensive survey of large model and agent safety. *arXiv preprint arXiv:2502.05206*, 2025.

Authorized licensed use limited to: University of Arkansas Litte Rock. Downloaded on August 10,2026 at 21:02:16 UTC from IEEE Xplore. Restrictions apply.